MONGODB

BEGINNER TO ADVANCED

DAY 1: INTRODUCTION TO MONGODB

What is MongoDB: Introduction to NoSQL databases, and how MongoDB differs from relational databases.

Document-Oriented: Understand MongoDB as a document-based database using JSON-like data.

Install MongoDB: Set up MongoDB on your machine or use MongoDB Atlas for a cloud-based solution.

Basic Commands: Start the MongoDB shell and learn basic commands like **show dbs, use,** and **show collections.**

DAY 2: INSERTING AND QUERYING DOCUMENTS

Documents: MongoDB stores data in BSON format (Binary JSON), and documents are the core data structure.

Insert Data: Learn how to insert single and multiple documents using **insertOne()** and **insertMany().**

Query Documents: Retrieve documents from a collection using **find().**

Comparison Operators: Use basic comparison operators like **$gt, $lt, $eq, $ne.**

DAY 3: UPDATING AND DELETING DOCUMENTS

Update Documents: Modify data using **updateOne()**, **updateMany()**, and **$set.**

Delete Documents: Remove documents using **deleteOne()** and **deleteMany().**

Replace Documents: Use **replaceOne()** to replace entire documents.

Update Operators: Learn operators like **$inc, $min, $max, and $rename.**

DAY 4: QUERYING WITH CONDITIONS AND LOGICAL OPERATORS

Advanced Queries: Learn how to use multiple conditions in queries.

Logical Operators: Use logical operators like **$and, $or, $not, $nor** for complex queries.

Element Queries: Query fields based on their presence (**$exists**) or data type (**$type**).

Example: Retrieve students who have scored between 70 and 90 or have not submitted a particular assignment.

DAY 5: INDEXING FOR PERFORMANCE

Indexes: Understand the purpose of indexes and how they improve query performance.

Create Indexes: Use **createIndex()** to add single-field and compound indexes.

Unique Indexes: Ensure uniqueness for certain fields using unique indexes.

Text Indexes: Learn how to create text indexes for searching string data.

Index Management: View and drop indexes using **getIndexes()** and **dropIndex()**.

DAY 6: AGGREGATION FRAMEWORK

Aggregation: Understand the MongoDB aggregation framework for advanced data processing.

Pipeline Stages: Learn about stages like **$match, $group, $sort, $project, $limit**, and **$skip**.

Aggregation Operators: Use operators like **$sum, $avg, $push, $first, $last,** and **$addToSet.**

$lookup: Perform joins between collections using **$lookup**.

DAY 7: RELATIONSHIPS AND EMBEDDING DOCUMENTS

Data Modeling: Understand how to model relationships in MongoDB using embedding or referencing.

Embedded Documents: Store related data inside a document (denormalization).

Referencing: Use references (normalization) to link data across collections.

Pros and Cons: Learn the trade-offs between embedding and referencing.